$$z \coloneqq \frac{1}{G}\sum_{g=1}^{G} E_g(X_g), \qquad \hat{X}_g \coloneqq D_g(z), \qquad \mathcal{L} \coloneqq \sum_{g=1}^{G} \|X_g - D_g(z)\|_2^2$$